Homework 4

MATH 441: DIFFERENTIAL EQUATIONS

NAME: DATE:

(Exercises are taken from *Elementary Differential Equations and Boundary Value Problems* by William E. Boyce and Richard C. DiPrima)

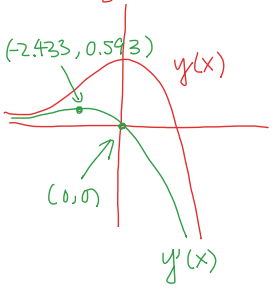
In each of Problems 1 and 2, find the solution of the given initial value problem. Sketch the graph of the solution and describe its behavior as t increases.

1. Exercise §3.1 #11.

$$6y'' - 5y' + y = 0, \quad y(0) = 4, \quad y'(0) = 0$$

$y(x)$ increases $(-\infty, -2.433)$, then decreases $(-2.433, \infty)$

$y(x)$ increases $(-\infty, -2.433)$ with increasing rate; $(-2.433, 0)$ with decreasing rate, then decreases with increasing rate



$y'' - \frac{5}{6}y' + \frac{1}{6}y = 0$

$y = e^{rx}$

$y' = re^{rx}$

$y'' = r^2e^{rx}$

$r^2e^{rx} - \frac{5}{6}re^{rx} + \frac{1}{6}e^{rx} = 0$

$r^2 - \frac{5}{6}r + \frac{1}{6} = 0$

$6r^2 - 5r + 1 = 0$

$(3r-1)(2r-1) = 0$

$r = \frac{1}{3}, \frac{1}{2}$

$y(t) = C_1e^{\frac{1}{3}x} + C_2e^{\frac{1}{2}x}$

$y'(t) = \frac{1}{3}C_1e^{\frac{1}{3}x} + \frac{1}{2}C_2e^{\frac{1}{2}x}$

$\boxed{y(t) = 12e^{\frac{1}{3}x} - 8e^{\frac{1}{2}x}}$

$y(0) = 4 = C_1 + C_2$

$4 = -\frac{1}{2}C_2$

$y'(0) = 0 = 2C_1 + 3C_2$

$-\frac{3}{2}C_2 = C_1$

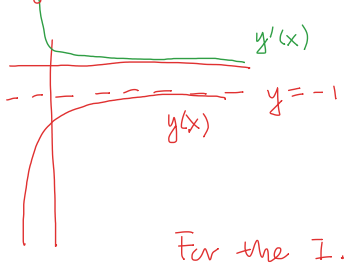
$C_2 = -8 \quad C_1 = 12$

2. Exercise §3.1 #12.

$$y'' + 3y' = 0, \quad y(0) = -2, \quad y'(0) = 3$$

$y'(x) > 0$, decreases $(-\infty, \infty)$

$y(x) < -1$, increases with decreasing rate $(-\infty, \infty)$ approaching asymptote $y = -1$



$y = e^{rx}$, $y' = re^{rx}$, $y'' = r^2e^{rx}$

$r^2e^{rx} + 3re^{rx} = 0$

$r^2 + 3r = 0$

$r = 0, -3$

$y(x) = C_1 + C_2e^{-3x}$

$y'(x) = -3C_2e^{-3x}$

$\boxed{y(x) = -1 - e^{-3x}}$

$y(0) = -2 = C_1 + C_2$

$C_1 = -1$

$y'(0) = 3 = -3C_2$

$C_2 = -1$

For the I.V.P. specifically, take graph description $x \geq 0$ for problems 1 & 2, since the question asked for increasing x with x starting at the initial value

In each of Problems 3 and 4, determine the linges interval in which the given intial value problem iscertain to have a unique twice-differentiable solution. Do not attempt to find the solution. Explain your answer.

3. Exercise §3.2 #10.

linear

$$y'' + (\cos t)y' + 3(\ln|t|)y = 0, \quad y(2) = 3, \quad y'(2) = 1$$

$p(t)$ $q(t)$ $g(t)$

$p(t) : (-\infty, \infty) \to \mathbb{R}$

$q(t) : (-\infty, 0) \cup (0, \infty) \to \mathbb{R}$

$g(t) : (-\infty, \infty) \to \mathbb{R}$

$t_0 = 2 \in (0, \infty)$

$\therefore \exists! y(t) \in C^2((0, \infty))$ such that y solves the I.V.P.

according to the Theorem of Existence & Uniqueness of 2nd order diff eqs.

4. Exercise §3.2 #11.

linear

$$(x-3)y'' + xy' + (\ln|x|)y = 0, \quad y(1) = 0, \quad y'(1) = 1$$

$$y'' + \frac{x}{x-3}y' + \frac{\ln|x|}{x-3}y = 0$$

$p(x)$ $q(x)$ $g(x)$

$p(x) : (-\infty, 3) \cup (3, \infty) \to \mathbb{R}$

$q(x) : (-\infty, 0) \cup (0, 3) \cup (3, \infty) \to \mathbb{R}$

$g(x) : (-\infty, \infty) \to \mathbb{R}$

$x_0 = 1 \in (0, 3)$

$\therefore \exists! y(x) \in C^2((0, 3))$ such that y solves the I.V.P.

due to the theorem of Existence & Uniqueness for 2nd order diff eqs.

5. Exercise §3.3 #19. Find the solution of the given initial value problem. Sketch the graph of the solution and describe its behavior for increasing t. sorry I used x, didn't see this initially

$$y'' - 2y' + 5y = 0, \qquad y(\pi/2) = 0, \qquad y'(\pi/2) = 2$$

$y(x) = c_1 e^{(1+2i)x} + c_2 e^{(1-2i)x}$

$= e^x \left(c_1 e^{2ix} + c_2 e^{-2ix} \right)$

$= e^x \left(c_1 (\cos 2x + i \sin 2x) + c_2 (\cos 2x - i \sin 2x) \right)$

$= e^x \left((c_1 + c_2) \cos 2x + (c_1 - c_2) i \sin 2x \right)$

let $A = c_1 + c_2$, $B = (c_1 - c_2) i$

$y(x) = e^x (A \cos 2x + B \sin 2x)$

$y(\frac{\pi}{2}) = 0 = -e^{\frac{\pi}{2}} A \quad A = 0 \quad \therefore \; y(x) = e^x B \sin 2x$

$y'(x) = Be^x \sin 2x + 2e^x \cos 2x) = Be^x (\sin 2x + 2\cos 2x)$

$y'(\frac{\pi}{2}) = 2 = -2Be^{\frac{\pi}{2}}$

$B = -e^{-\frac{\pi}{2}}$

$y(x) = -e^{x - \frac{\pi}{2}} \sin 2x$

$r^2 e^{rx} - 2r e^{rx} + 5e^{rx} = 0$

$r^2 - 2r + 5 = 0$

$\frac{2 \pm \sqrt{4 - 20}}{2} = r$

$1 \pm 2i = r$

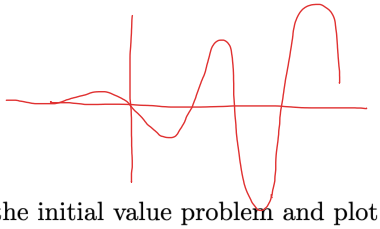
graph oscillates with period π & increasing amplitude similar to a sin wave with $y(0) = 0$

$\text{Amp} = e^{x - \frac{\pi}{2}}$

$\lim_{x \to \infty} \text{Amp} = \infty$

$B = 2$

$T = \frac{2\pi}{B} = \pi$



6. Exercise §3.4 #15(a). Solve the initial value problem and plot its solution for $0 \le t \le 5$.

$$4y'' + 12y' + 9y = 0, \qquad y(0) = 1, \qquad y'(0) = -4$$

sorry I used x, didn't see this initially

$y_1(x) = Ce^{-\frac{3}{2}x}$

$y(x) = v(x) e^{-\frac{3}{2}x}$

$y'(x) = v'(x) e^{-\frac{3}{2}x} - \frac{3}{2} v(x) e^{-\frac{3}{2}x}$

$y''(x) = v''(x) e^{-\frac{3}{2}x} - \frac{3}{2} v'(x) e^{-\frac{3}{2}x} - \frac{3}{2} v'(x) e^{-\frac{3}{2}x} + \frac{9}{4} v(x) e^{-\frac{3}{2}x}$

$= v''(x) e^{-\frac{3}{2}x} - 3v'(x) e^{-\frac{3}{2}x} + \frac{9}{4} v(x) e^{-\frac{3}{2}x}$

$y'' + 3y' + \frac{9}{4} y = 0$

$r^2 + 3r + \frac{9}{4} = 0$

$4r^2 + 12r + 9 = 0$

$(2r + 3)^2 = 0$

$r = -\frac{3}{2}$

$4e^{-\frac{3}{2}x} (v''(x) - 3v'(x) + \frac{9}{4} v(x)) + 12e^{-\frac{3}{2}x} (v'(x) - \frac{3}{2} v(x)) + 9v(x) e^{-\frac{3}{2}x} = 0$

$4v''(x) = 0$

$v''(x) = 0$

$v'(x) = c_1$

$v(x) = c_1 x + c_2$

$y(x) = c_1 x e^{-\frac{3}{2}x} + c_2 e^{-\frac{3}{2}x} \qquad y(0) = 1 = c_2$

$y'(x) = (c_1 - \frac{3}{2}) e^{-\frac{3}{2}x} - \frac{3}{2} c_1 x e^{-\frac{3}{2}x} \qquad y'(0) = -4 = c_1 - \frac{3}{2}$

$c_1 = -\frac{5}{2}$

$y(x) = -\frac{5}{2} x e^{-\frac{3}{2}x} + e^{-\frac{3}{2}x}$

